

DATA INTEGRITY REPORT

1. Missing Values

- Column: return_date and publication_year
- Affected Records: Checkouts from the Reading Kickoff web page lack return dates, and specific book catalog entries lack publication years.
- Decision & Reasoning: For return dates, filled missing values with "Not Due Yet" because kickoff items have extended summer terms. For publication years, filled missing values with "0" to keep data types consistent without deleting valid inventory rows.

2. Duplicate Records

- Column: Entire row dataset
- Affected Records: Exact duplicated checkout lines resulting from multiple join combinations.
- Decision & Reasoning: Applied standard deduplication to drop exact structural clones while keeping records that were only similar, ensuring no valid distinct user events were accidentally deleted.

3. Inconsistent Text Values

- Column: neighborhood and membership_status
- Affected Records: Text entries containing mixed capitalization and extra spacing (e.g., "zamalek ", "Active", "active").
- Decision & Reasoning: Converted all text strings to uniform title case and stripped trailing spaces so grouping and aggregation functions categorize neighborhoods and statuses accurately.

4. Checkouts With No Matching Member

- Column: member_id
- Affected Records: Orphan checkouts pointing to an ID not registered in the members table.
- Decision & Reasoning: Removed these unmatched orphan checkouts completely to prevent ghost data from skewing user metrics and final neighborhood fairness calculations.